

## Experiment No. – 6

### **Aim –**

(A.) A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day, calculated as  $\text{Difference} = \text{Apples Sold} - \text{Oranges Sold}$ . Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

(B.) A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where id represents the current node and p\_id represents the parent node. Classify every node into one of the following categories: Root — a node that does not have any parent; Leaf — a node that has a parent but does not have any children; Inner — a node that has both a parent and at least one child. Display the node ID along with its category, sorted by node ID.

### **Tools Required**

- Postgres
- PgAdmin4

### **Queries:**

A.

```
CREATE TABLE Sales(  
    sale_date DATE,  
    fruit VARCHAR(20),  
    sold_num INT,  
    PRIMARY KEY (sale_date, fruit)  
);
```

```
INSERT INTO Sales  
VALUES  
('2020-05-01','apples',10),  
('2020-05-01','oranges',8),
```

```
('2020-05-02','apples',15),
('2020-05-02','oranges',15),

('2020-05-03','apples',20),
('2020-05-03','oranges',0),

('2020-05-04','apples',15),
('2020-05-04','oranges',16);

select sale_date , sum(
    CASE
        WHEN fruit = 'apples' then sold_num
        Else -sold_num
    end
) as diff
from Sales
group by sale_date
Order by sale_date;REFERENCES Signups(user_id)
);
```

**B.**

```
SELECT ID,
    CASE
        WHEN P_ID IS NULL THEN 'Root'
        WHEN ID IN (SELECT DISTINCT P_ID FROM TREE WHERE P_ID IS NOT
        NULL) THEN 'Inner'
        ELSE 'Leaf'
    END AS TYPE
FROM TREE ORDER BY ID;
```

## Outputs:

|   | sale_date<br>date | diff<br>bigint |
|---|-------------------|----------------|
| 1 | 2020-05-01        | 2              |
| 2 | 2020-05-02        | 0              |
| 3 | 2020-05-03        | 20             |
| 4 | 2020-05-04        | -1             |

## Learning Outcome

After performing this SQL query, the following learning outcomes were achieved:

1. **Learned to create and populate a database table** using CREATE TABLE and INSERT INTO statements.
2. **Understood the use of a composite primary key** (sale\_date, fruit) to uniquely identify each sales record.
3. **Learned to use the CASE statement** for applying conditional logic within an SQL query.
4. **Used the SUM() aggregate function** to calculate the difference between the number of apples and oranges sold on each date.
5. **Understood the use of GROUP BY** to perform calculations separately for every sale\_date.
6. **Learned to use ORDER BY** to display the results in chronological order.